8) Rank(AB) = rank(B) for an invertible A $(n \times n)$ & B $(n \times m)$.

Write $B = [v_1\ v_2 \cdots v_m]$; v_i column vector

$\Rightarrow AB = [Av_1, Av_2 \cdots Av_m]$

Then rank(AB) = rank B

9) What are the eigenvalues of a nilpotent matrix?

If $A^k = 0$ (ie nilpotent)

& $Av = \lambda v$, $v \neq 0$,

then $A^k v = \lambda^k v$

$\Rightarrow \lambda^k v = 0 \Rightarrow \lambda = 0$, is the only eigenvalue of A & eigenvector is $A^{k-1}(v)$ if $A^{k-1} \neq 0$

(6) If $T: V_1 \to V_2$ is injective linear transformation show that $x=0$ is the only vector such that $T(x)=0$.

Pf: Suppose T is injective, i.e. 1-1

$\Leftrightarrow$ $Tv = Tu$ possible for $v=u$ only

$\Leftrightarrow$ $x=0$ is the only ~~solu~~ vector such that $Tx=0$ ~~else $T(0)=0$~~

(7) Show that a nilpotent matrix A is diagonalizable $\Leftrightarrow$ $A=0$.

If $A^k=0$ for some k & A is diagonalizable, then $A = QDQ^{-1}$ for some Q & D diagonal matrix.

$\Rightarrow A^k = QD^kQ^{-1} = 0$

$\Rightarrow D^k = 0 \Rightarrow D=0 \Rightarrow A=0$

(3) (Contd). The answer is

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(4) Find the eigenvalues of the matrix $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Hence find A^{-1}

$\lambda^2 - 1 = 0$; $\lambda = \pm 1$

Cayley-Hamilton $\Rightarrow$ $A^2 - I = 0$

$A - A^{-1} = 0$

$A^{-1} = A$

(5) Suppose $AX = b$ is a system, where A is $m \times n$, X is $n \times 1$ & b is $m \times 1$. If rank $A \leq \frac{n}{2}$ can the unique solution exist?

~~Yes~~ NO ;

For unique soln, rank A = rank $\tilde{A} = n$

(3) Eg Check if Consistent

$$x + 2y = 1$$
$$2x + 3y = 1$$
$$x + y = 0$$

Check whether it is consistent

$$\left[\begin{array}{cc|c} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 0 \end{array}\right] \sim \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right]$$

A $\quad$ $\tilde{A}$

rank $A = 2$

rank $\tilde{A} = 2$, = the no. of variables

$\Rightarrow$ ~~Contradict~~ Consistent & unique

$$\begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$x + 2y = 1$

$$\boxed{\begin{array}{l} y = 1 \\ x = -1 \end{array}}$$

unique soln

Find the matrix of the linear map

$T : \mathbb{R}^3 \longrightarrow \mathbb{P}_3$ w.r.to basis

$\{e_1, e_2, e_3\}$ & $\{v_1, v_2, v_3, v_4\}$

$v_1 = x$

$v_2 = 1$

$v_3 = 1 + x^2$

$v_4 = 1 + x^3$

$T(a, b, c) = a + bx^2$

4×3

To: $a + bx^2 = \alpha_1 x + \alpha_2 + \alpha_3 (1 + x^2) + \alpha_4 (1 + x^3)$

$\Rightarrow \cancel{\alpha_2 + \alpha_3 + \alpha_4 = 0}$

$\Rightarrow \alpha_2 + \alpha_3 + \alpha_4 = a$

$\alpha_1 = 0$

$\alpha_3 = b$; $\alpha_4 = 0$

$\alpha_1 = 0, \quad \alpha_3 = b, \quad \alpha_4 = 0, \quad \alpha_2 = a - b.$

$\Rightarrow T(a, b, c) = a + bx^2$

$= 0 \cdot x + (a - b) \cdot 1 + b(1 + x^2) + 0(1 + x^3)$

$Te_1 = 0 \cdot x + 1 \cdot 1 + 0(1 + x^2) + 0(1 + x^3)$, $Te_2 = 0 \cdot x + 0 \cdot 1 + 0(1 + x^2) + 0(1 + x^3)$

$\Rightarrow Te_3 = 0 \cdot x + 0 \cdot 1 + 0(1 + x^2) + 0(1 + x^3)$

Find the a basis of the de

(1)

$$V = \left\{ \begin{pmatrix} x_1 + x_2 \\ 2x_1 - x_2 + x_3 \\ 2x_2 - 3x_3 \end{pmatrix} \right\} \text{ of } \mathbb{R}^2$$

$$= \left\{ x_1 \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + x_3 \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} \right\}$$

$= \text{Span}\{v_1, v_2, v_3\}$, where

$v_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$, $v_4 = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}$

~~If~~ clear that v_1, v_2 are L.I

$x_1 + x_2 = 0$

$2x_1 - x_2 = x_3$

$2x_2 = 3x_3$

$x_1 = x_3$

$-2x_1 = 3x_3$

$$\begin{bmatrix} 1 & 2 & 0 \\ 1 & -1 & 2 \\ 0 & 1 & -3 \end{bmatrix}$$

L.I.

echelon $\begin{bmatrix} = \\ = \\ = \end{bmatrix}$ non zero rows $\Rightarrow \{v_1, v_2, v_3\}$

You will get 3 basis for Y.